CI Outcome Prediction Project

Supplementary Information

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1 Supplementary Figures

This appendix contains additional figures supporting the main analyses.

1.1 WRS Improvement Distribution

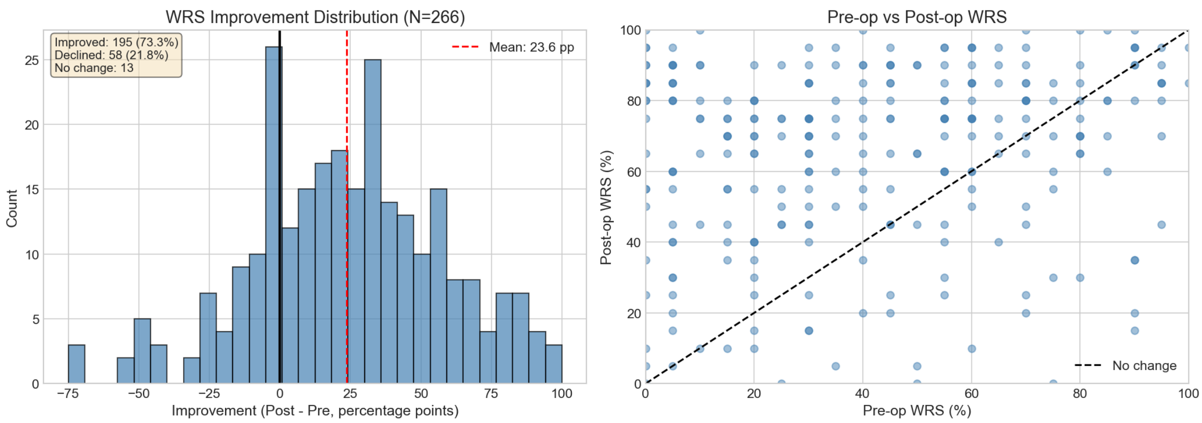

Figure 1: Left: Distribution of WRS improvement (post–pre) among patients with both measurements (N=266). Mean improvement is 23.6 percentage points. 73.3% of patients improved, 21.8% declined, and 4.9% showed no change. Right: Pre-operative vs. post-operative WRS showing individual trajectories; points above diagonal indicate improvement.

1.2 Speech Test Distributions

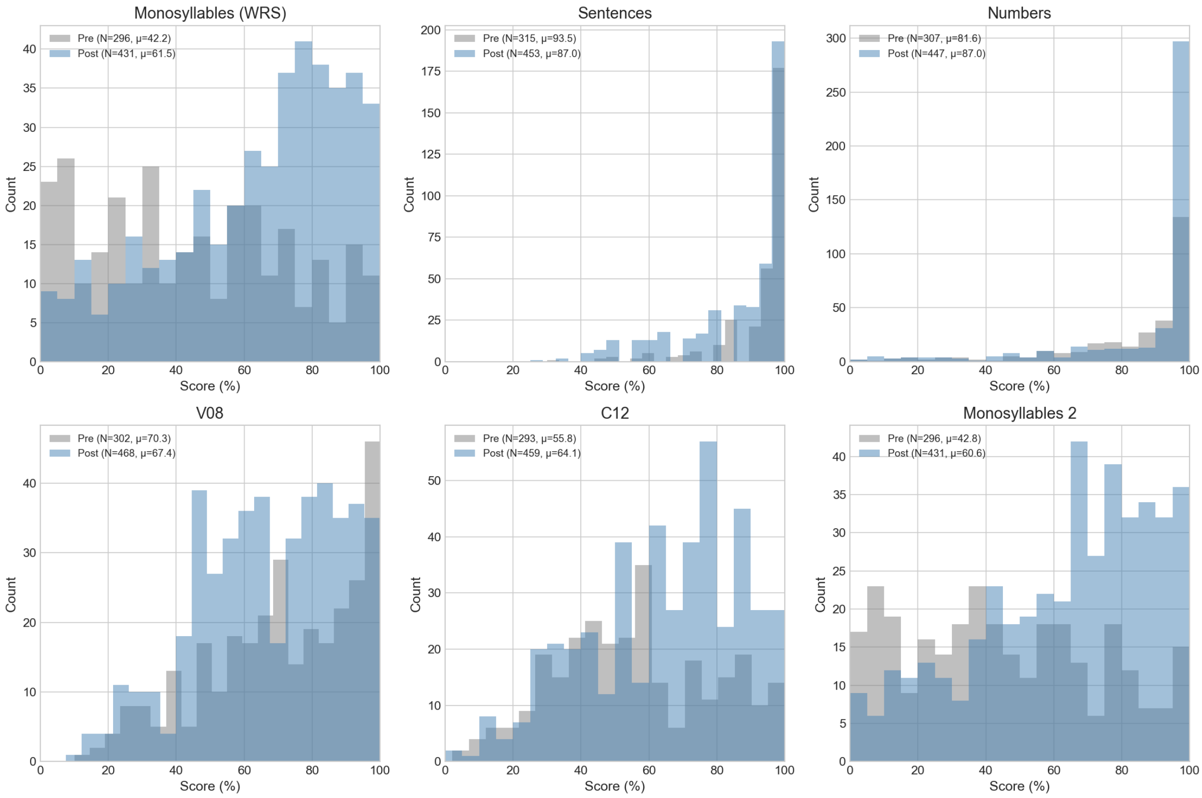

Figure 2: Pre-operative (gray) and post-operative (blue) distributions for all speech tests. Monosyllables (WRS) is the primary outcome used in this study. Sentences and numbers show ceiling effects. V08 (OLSA) shows minimal change post-CI. C12 (consonants) shows moderate improvement.

1.3 Audiometric Data Quality

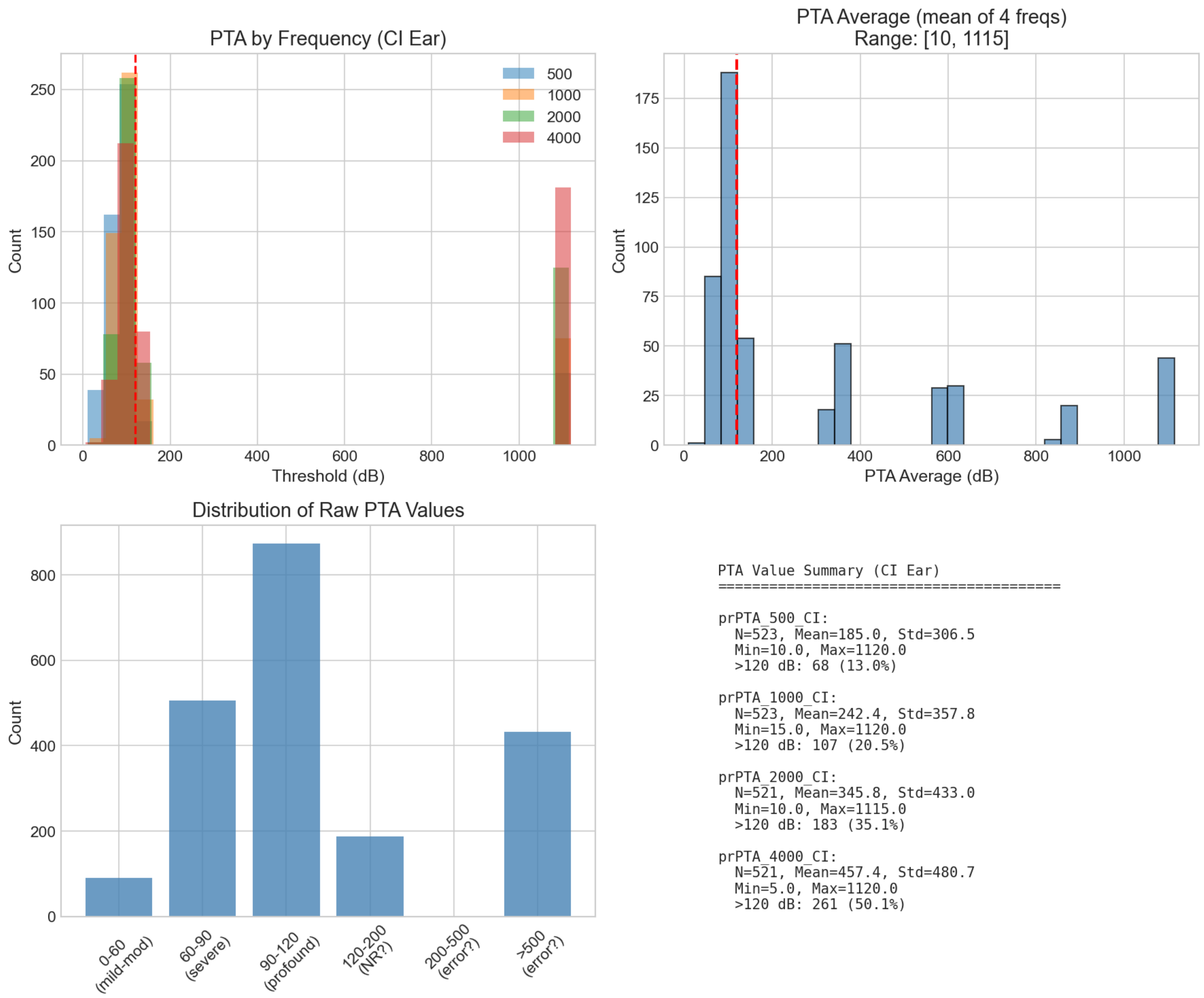

Figure 3: Pure-tone audiometry data quality diagnostics. Top left: PTA by frequency showing expected severe-profound hearing loss in CI ear. Top right and bottom: Raw PTA values reveal coding inconsistencies—values >120 dB (up to 1120 dB) represent "no response" coded numerically rather than as missing. These values are capped at 120 dB following Bertschinger's methodology.

1.4 Age Distribution: Adults vs. Children

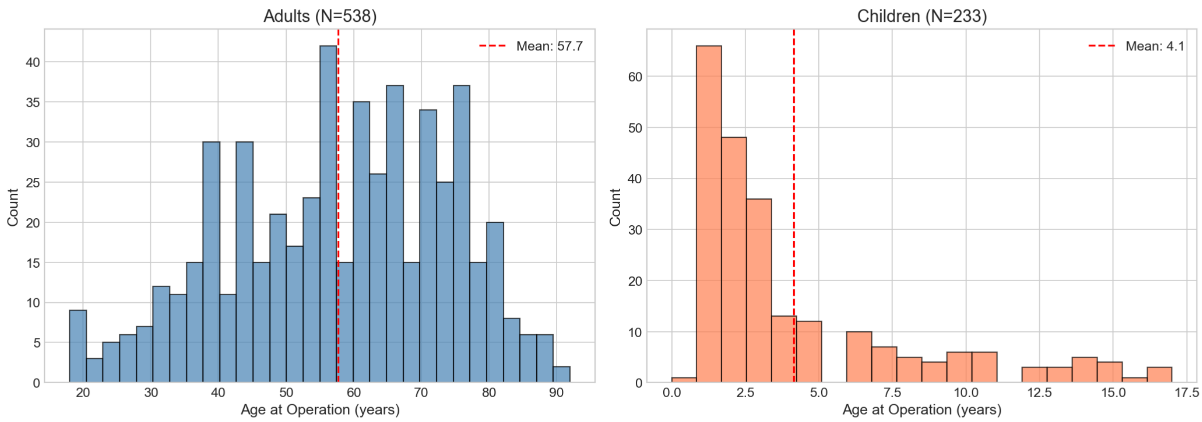

Figure 4: Age distribution at implantation for adults (N=538, mean 57.7 years) and children (N=233, mean 4.1 years). Adults show broad distribution from 18–93 years. Children are concentrated in early years with peak at 1–2 years, reflecting early intervention protocols for congenital hearing loss.

1.5 Pediatric WRS Availability

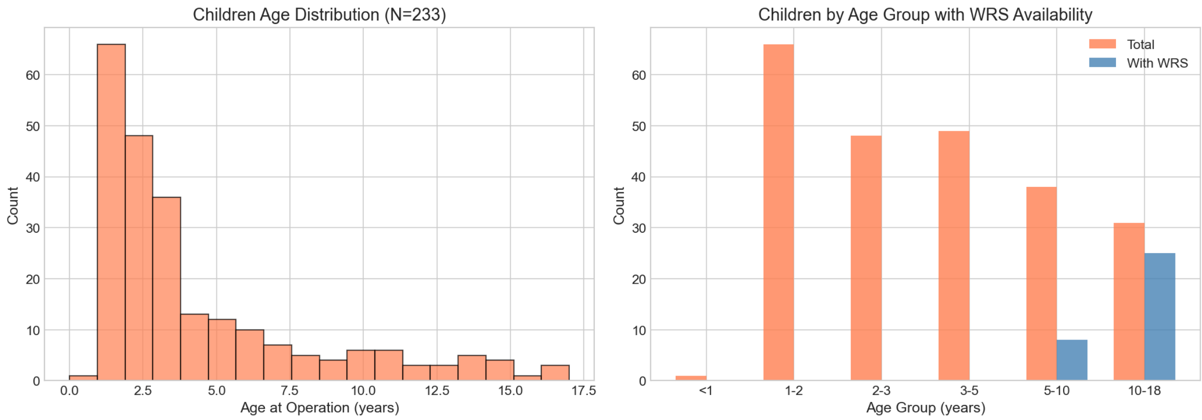

Figure 5: Left: Children age distribution showing concentration below 5 years. Right: WRS availability by age group. Children under 5 cannot perform word recognition tests, resulting in missing outcomes for 70% of pediatric patients. This motivates the need for CAP/SIR scores as alternative outcomes.

1.6 Temporal Distribution by Population

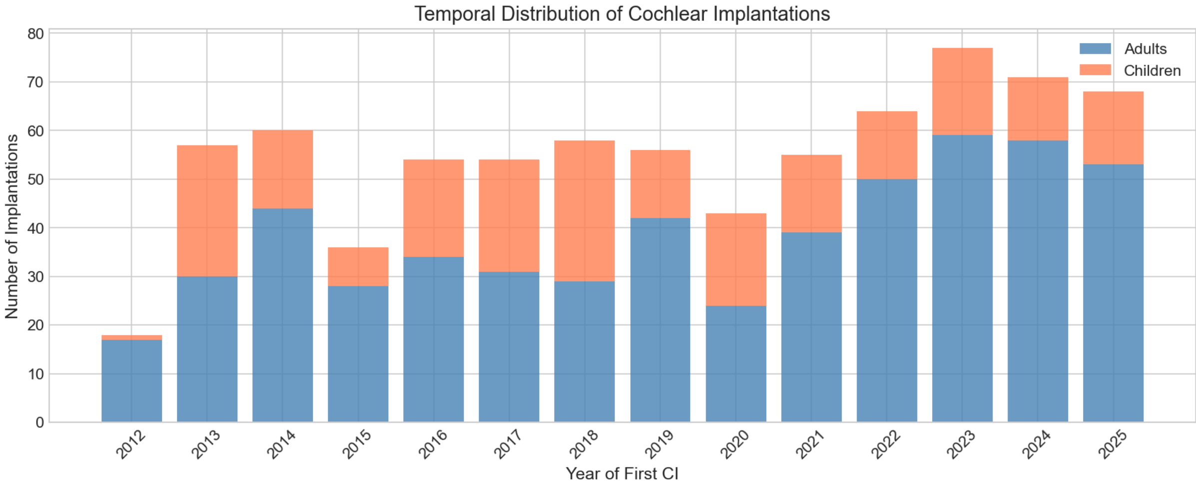

Figure 6: Temporal distribution of cochlear implantations by year, stratified by adults (blue) and children (orange). Note COVID-19 dip in 2020 and growth from 2022–2024. The proportion of pediatric implants has remained relatively stable over time.

2 Prediction Analysis Details

This appendix provides supplementary figures for the prediction analysis (RQ2) summarized in Section 3.

2.1 Coefficient Comparison with Bertschinger

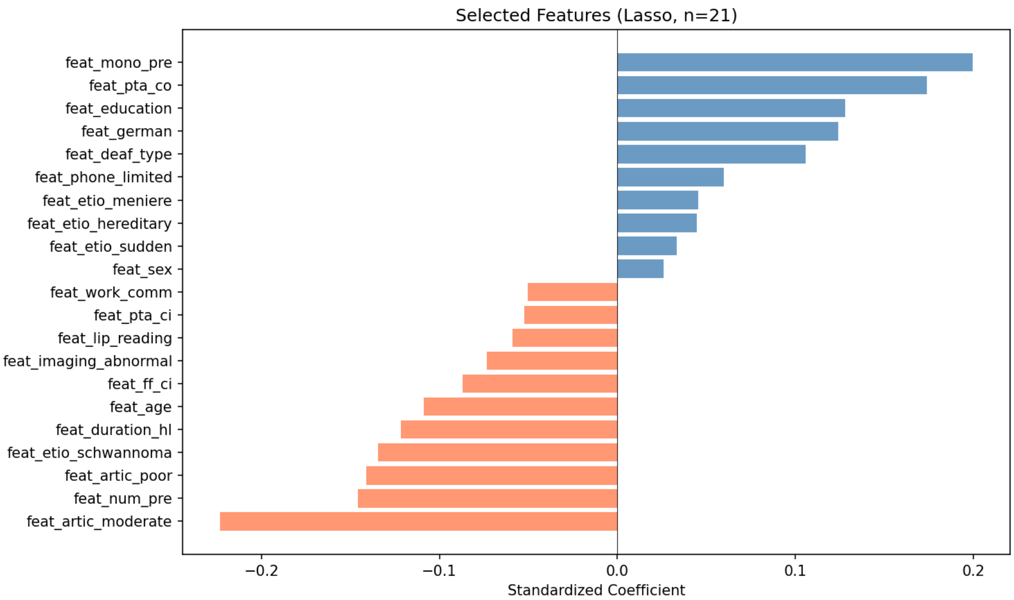

Figure 7: Standardized coefficients for Bertschinger's 7 features. Our coefficients (blue) are smaller in magnitude but all signs match Bertschinger's (green), confirming consistent predictor–outcome relationships despite different samples.

Feature	Our Coef.	Bertschinger	Sign Match?
Articulation (moderate)	−0.261	−0.624	✓
Articulation (poor)	−0.146	−0.824	✓
German speaker	+0.097	+0.500	✓
Phone use (limited)	+0.063	+0.264	✓
Etiology: Schwannoma	−0.154	−1.054	✓
Duration of HL (years)	−0.116	−0.014	✓
Free-field CI (dB)	−0.145	−0.022	✓

Table 1: Coefficient comparison using Bertschinger's exact 7 features. All 7 coefficient signs match, confirming the model captures the same relationships.

2.2 Feature Importance

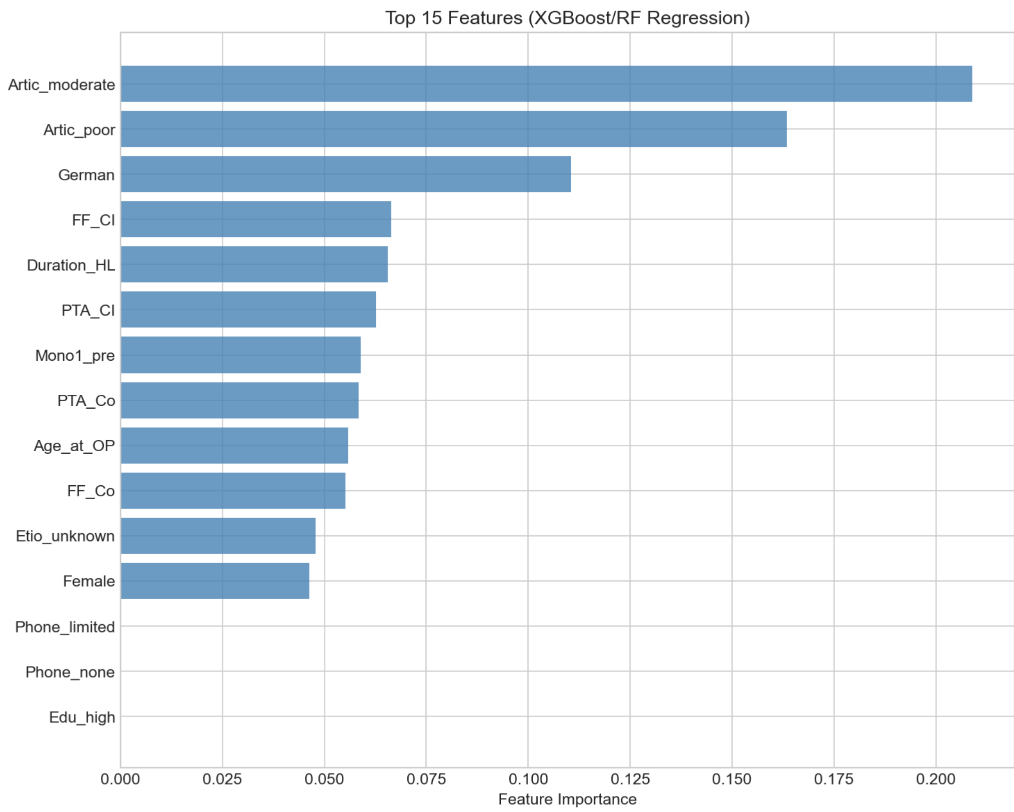

Figure 8: Feature importance from XGBoost model. Top predictors align with Bertschinger's selected features: articulation quality, German speaker status, free-field thresholds, and duration of hearing loss.

2.3 Predictor–Outcome Relationships

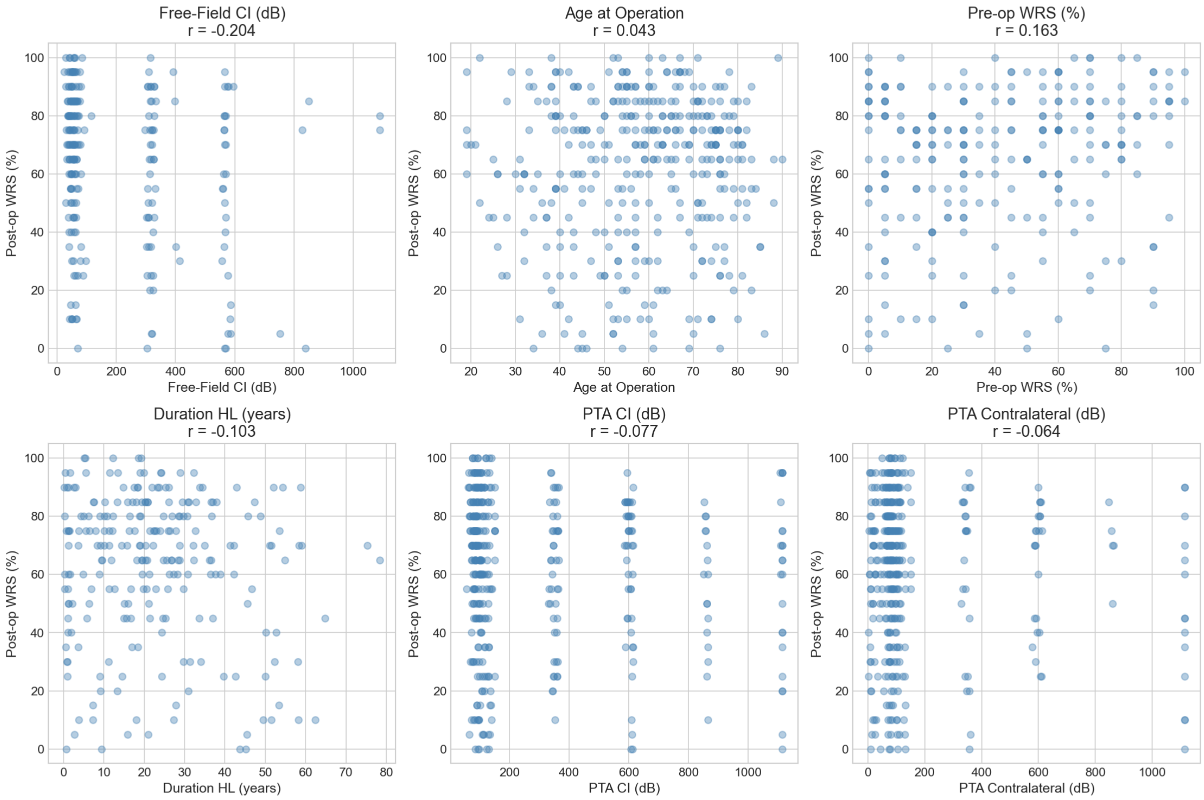

Figure 9: Bivariate relationships between continuous predictors and post-operative WRS. Correlations are weak to moderate, with free-field CI threshold showing strongest association (r = –0.20).

2.4 Lasso Feature Selection Details

Feature	Lasso Coefficient
Articulation	–0.182
Education	+0.099
FF 4000 Hz contralateral	–0.099
Articulation moderate	–0.096
German speaker	+0.090
PTA 2000 Hz contralateral	+0.087
Schwannoma	–0.083
FF 500 Hz contralateral	+0.082
FF 2000 Hz CI ear	–0.078
PTA 500 Hz CI ear	–0.061

Table 2: Top 10 Lasso-selected features by coefficient magnitude. Optimal regularization $\alpha = 0.025$ selected via cross-validation.

Comparison with Bertschinger features:

- 5/7 Bertschinger features retained: articulation, German, schwannoma, phone use, duration

- 2 features dropped: artic_poor (collinear with articulation), ff_ci (replaced by specific frequencies)
- New features added: education, lip reading, MRI findings, individual audiometric frequencies

3 Sensitivity Analysis Details

This appendix provides detailed figures for the sensitivity analyses summarized in Section 5.

3.1 Imputation Method Comparison

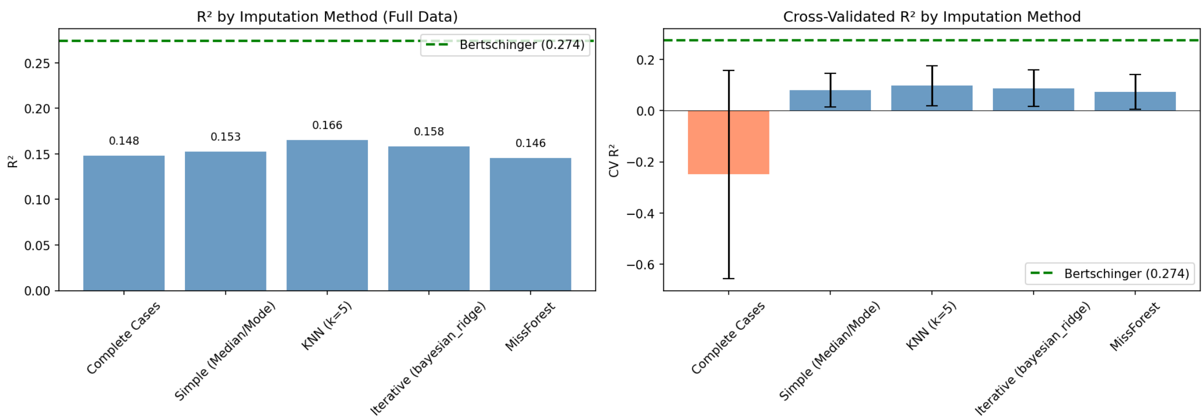

Figure 10: R^2 comparison across imputation methods. Left: Training R^2. Right: Cross-validated R^2 with error bars. Green dashed line shows Bertschinger's reported R^2=0.27. Complete cases analysis (N=145) shows high variance due to small sample size.

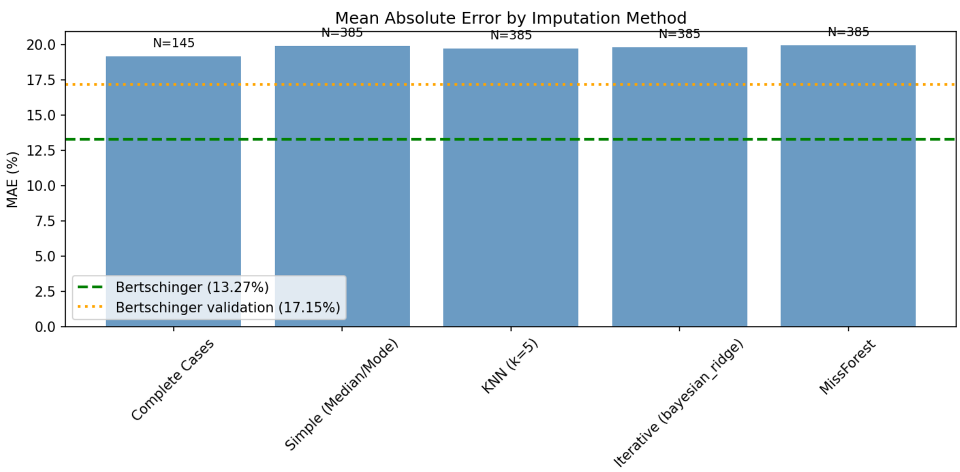

Figure 11: Mean Absolute Error by imputation method. All methods achieve MAE ≈19–20%, compared to Bertschinger's training MAE of 13.3% and validation MAE of 17.2%.

3.2 Coefficient Stability Across Imputation Methods

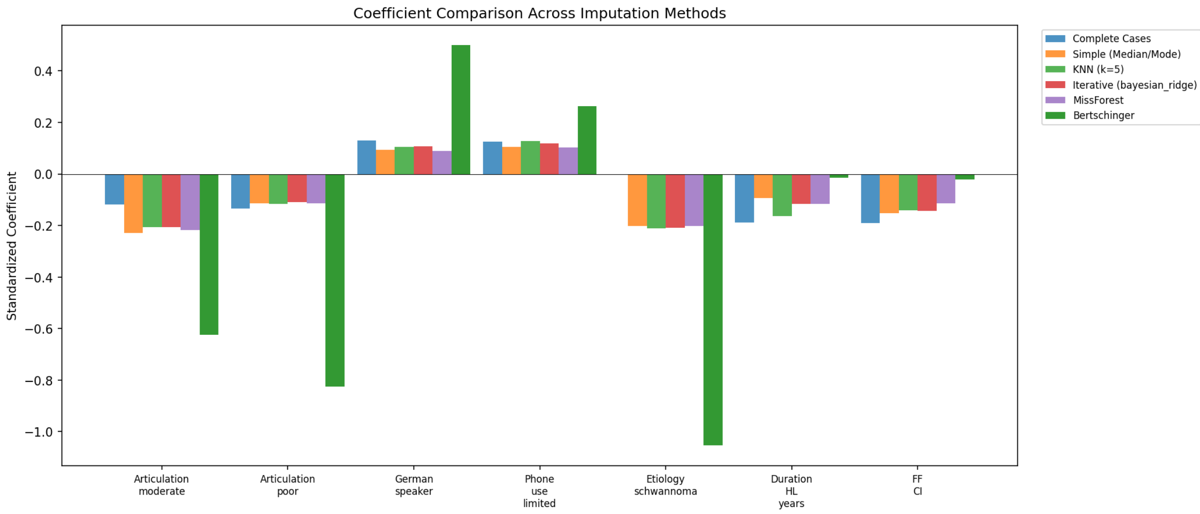

Figure 12: Coefficient estimates for Bertschinger's 7 features across imputation methods. Our coefficients (blue shades) are consistently smaller in magnitude than Bertschinger's (dark green), but direction is consistent across all methods.

Feature	KNN	Iterative	Simple	MissForest	Bertschinger
Articulation (moderate)	–0.26	–0.25	–0.24	–0.26	–0.62
Articulation (poor)	–0.15	–0.14	–0.13	–0.15	–0.82
German speaker	+0.10	+0.09	+0.08	+0.10	+0.50
Phone use (limited)	+0.06	+0.06	+0.05	+0.06	+0.26
Etiology: Schwannoma	–0.15	–0.16	–0.15	–0.15	–1.05
Duration of HL (years)	–0.12	–0.11	–0.10	–0.12	–0.01
Free-field CI (dB)	–0.15	–0.14	–0.13	–0.15	–0.02

Table 3: Standardized coefficient estimates across imputation methods. All signs match Bertschinger's original coefficients, confirming robust predictor–outcome relationships.

4 Clustering Algorithm Details

This appendix provides detailed methodology and algorithm comparison for the phenotype discovery analysis summarized in Section 6.

4.1 Algorithm Selection

Three clustering algorithms were compared: K-means (partition-based), Hierarchical with Ward linkage (agglomerative), and Gaussian Mixture Model (probabilistic). The optimal number of clusters was determined using silhouette analysis.

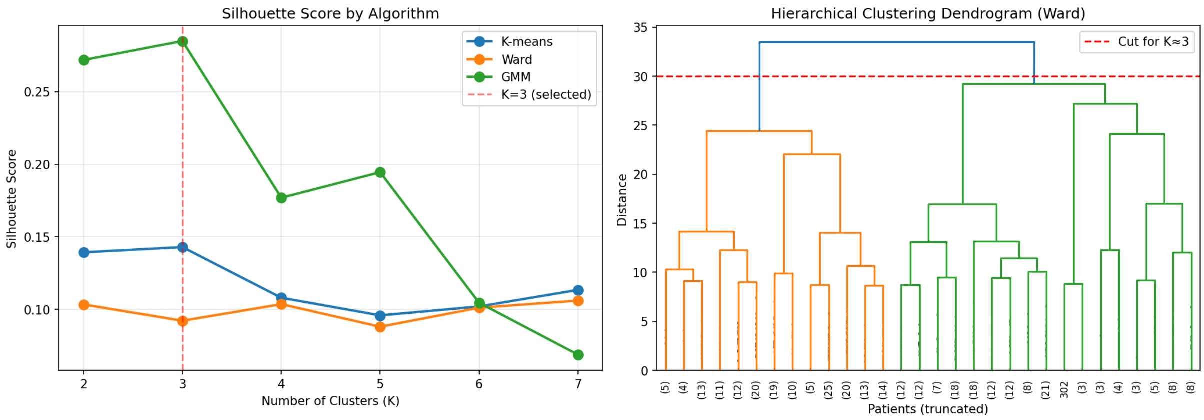

Figure 13: Left: Silhouette scores by number of clusters for each algorithm. K-means and GMM peak at K=3; Ward continues improving. Silhouette scores are low (<0.15) indicating weak cluster structure. Right: Hierarchical dendrogram showing patient groupings.

Algorithm	**K=2**	**K=3**	**K=4**	**Best K**
K-means	0.140	**0.141**	0.109	3
Hierarchical (Ward)	0.098	0.116	0.127	7
GMM	0.273	**0.285**	0.259	3

Table 4: Silhouette scores by algorithm and number of clusters. All scores are low (<0.30), indicating weak cluster structure.

4.2 Algorithm Comparison

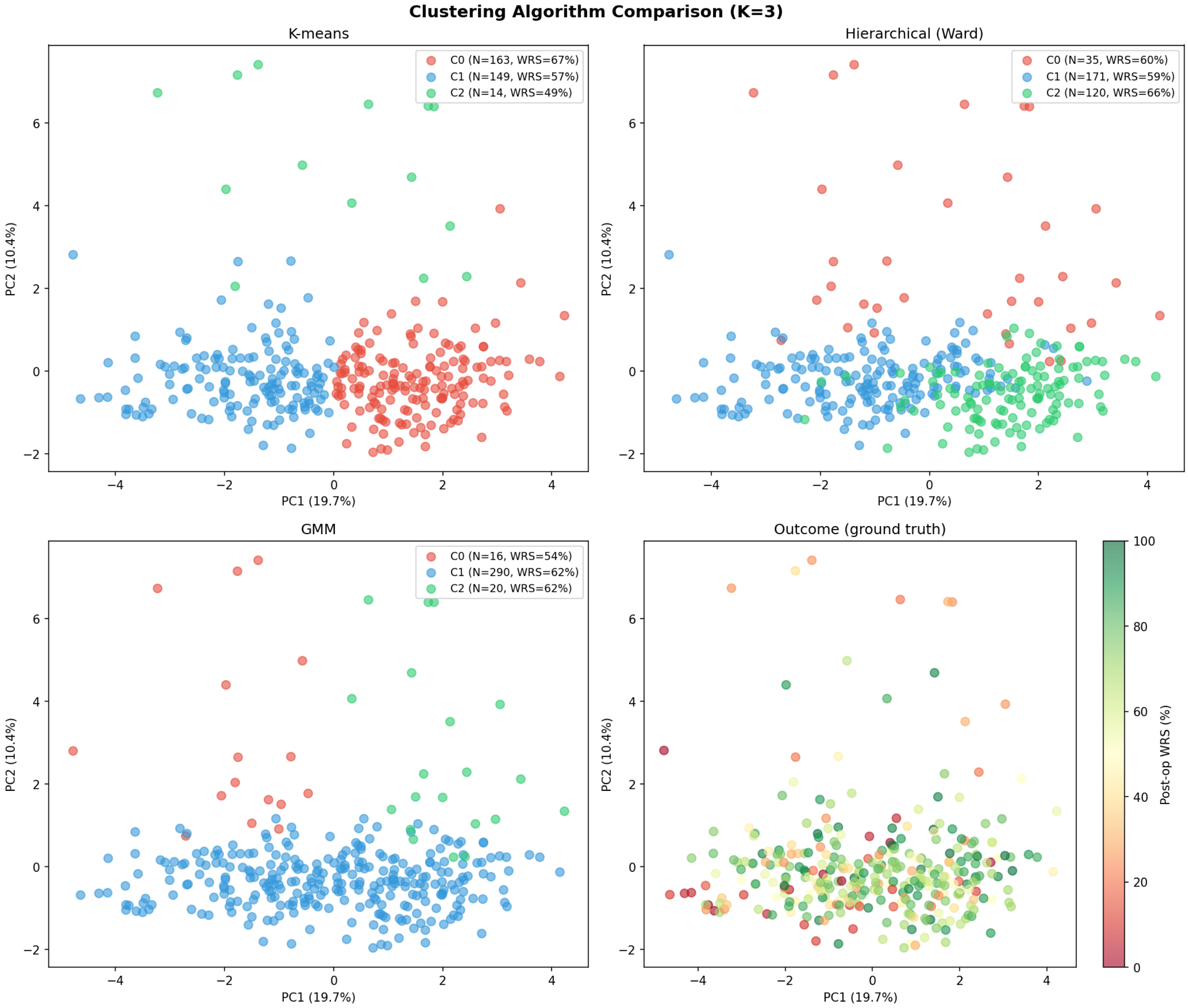

Figure 14: Clustering results in PCA space (30% variance explained). Top row: K-means and Hierarchical (Ward) show similar groupings. Bottom left: GMM finds different structure. Bottom right: True outcome distribution shows gradient rather than discrete groups.

Algorithm	Silhouette	Cluster Sizes	ANOVA F	p-value	Note
K-means	0.141	15, 148, 163	8.02	**0.0004**	Best outcome separation
Hierarchical (Ward)	0.116	36, 103, 187	3.10	0.046	More balanced sizes
GMM	0.285	290, 20, 16	0.79	0.453	Poor outcome separation

Table 5: Algorithm comparison (K=3). K-means achieves best outcome differentiation despite similar silhouette scores.

4.3 Cluster Agreement

Adjusted Rand Index (ARI) measures how often algorithms assign patients to the same cluster (1.0 = perfect agreement, 0 = random):

	K-means	Ward	GMM
K-means	1.00	0.47	0.13
Hierarchical (Ward)	0.47	1.00	0.31
GMM	0.13	0.31	1.00

Table 6: Cluster agreement (Adjusted Rand Index). K-means and Ward show moderate agreement (ARI=0.47). GMM finds substantially different structure.

4.4 Cluster Characterization

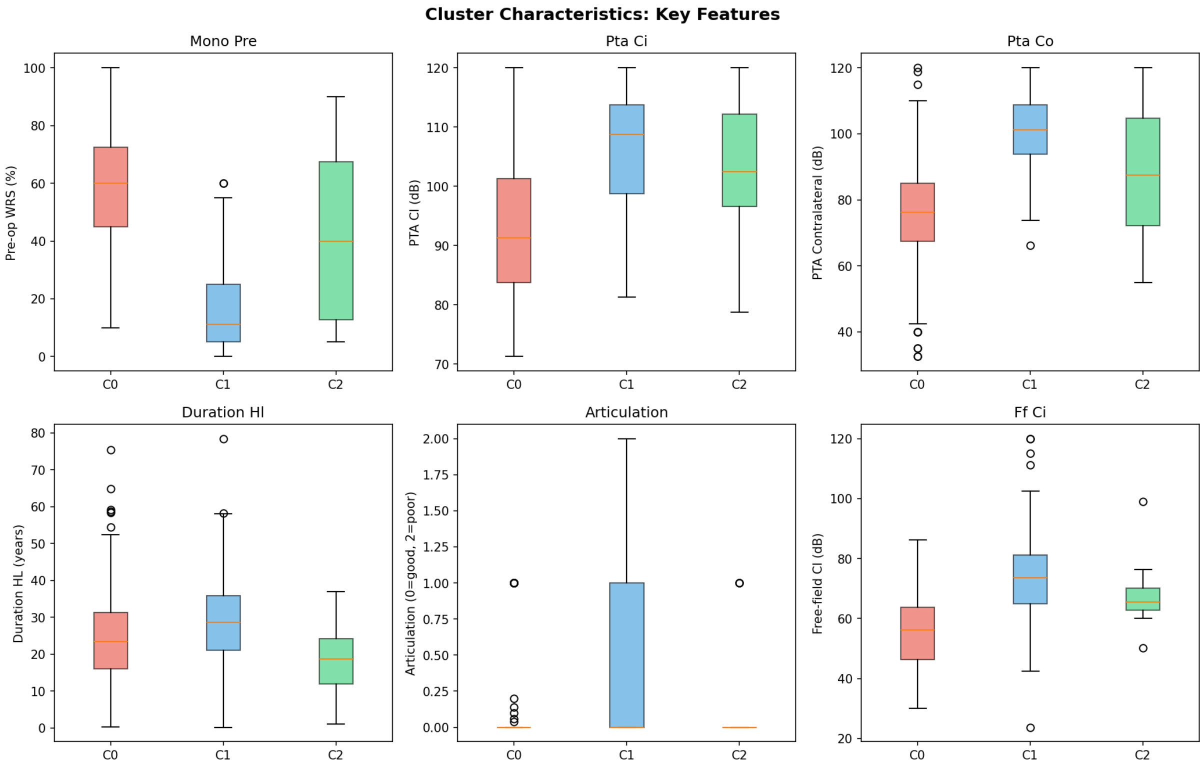

Figure 15: Key features distinguishing clusters. Pre-operative WRS shows the largest difference: Cluster 2 (good outcome) has highest pre-op WRS, while Cluster 1 (moderate) has lowest. Cluster 0 (poor outcome) is characterized by imaging abnormalities.

Feature	C0 (Poor)	C1 (Moderate)	C2 (Good)	Pattern
Pre-op WRS (%)	37.4	15.9	58.8	C2 > C0 > C1
PTA CI (dB)	104.8	106.2	93.3	C2 has better hearing
PTA Contralateral (dB)	90.0	100.9	75.4	C2 has more asymmetry
Duration HL (years)	18.6	29.1	24.4	C1 has longest duration
Articulation (0–2)	0.27	0.51	0.10	C2 has best articulation
MRI abnormal (%)	100%	0%	0%	C0 all have MRI findings
CT abnormal (%)	53%	5%	5%	C0 has CT abnormalities

Table 7: Cluster centers for key distinguishing features.

4.5 Per-Cluster Prediction

Model	N	CV R^2	vs. Global
Global (all patients)	326	0.102 ± 0.10	–
Cluster 0 only	15	–	Too small
Cluster 1 only	148	0.080 ± 0.11	–0.022
Cluster 2 only	163	0.001 ± 0.12	–0.101

Table 8: Per-cluster prediction performance. Fitting separate models per cluster does not improve R^2.

5 Missingness Analysis

This appendix details missing data patterns in both datasets.

5.1 Availability by Variable Category

Category	N Variables	Adults Avail.	Full Avail.
Demographics	9	83.5%	84.7%
Speech	18	70.6%	54.6%
Audiometry	40	81.8%	65.9%
Communication	16	72.9%	69.6%
Evaluation	27	66.0%	64.6%
Socioeconomic	36	54.5%	55.1%
CT Imaging	11	66.3%	64.6%
MRI Imaging	6	82.8%	81.6%
Surgical	20	74.0%	71.1%
Postoperative	12	52.6%	51.8%
Other	33	75.2%	73.4%
Total	**226**	**70.3%**	**65.3%**

Table 9: Variable availability by category. Adults dataset (N=479) vs Full dataset including children (N=771).

5.2 Key Predictor Availability

Variable	Adults N	Adults %	Full N	Full %
Age_at_OP	479	100.0%	771	100.0%
PTA_CI	473	98.7%	704	91.3%
PTA_Co	473	98.7%	705	91.4%
COM_ARTICULATION	462	96.5%	721	93.5%
COM_MAIN_LANGUAGE	472	98.5%	762	98.8%
EVA_ETIOLOGY	479	100.0%	771	100.0%
Mono1_pre	280	58.5%	307	39.8%
Mono1_post	385	80.4%	464	60.2%

Table 10: Availability of key predictors (Bertschinger features) across datasets.

5.3 Outcome Missingness Pattern

	Has Outcome	Missing Outcome
N	464 (60.2%)	307 (39.8%)
Mean age	54.9 years	21.4 years
Children <5	0	164
Adults $\geq$18	431	107

Table 11: Outcome missingness is strongly age-related: all children <5 are missing speech outcomes (MNAR pattern).

5.4 Complete Cases Analysis

Features	Adults N (%)	Full N (%)
Age only	479 (100.0%)	771 (100.0%)
+ Audiometry	472 (98.5%)	696 (90.3%)
+ Communication	450 (93.9%)	648 (84.0%)
+ Pre-op WRS	272 (56.8%)	297 (38.5%)
+ Outcome (complete)	237 (49.5%)	264 (34.2%)

Table 12: Cumulative complete cases as features are added. Adding Mono1_pre and outcome substantially reduces available N.

5.5 Implications

1. **MNAR pattern:** Outcome missingness is age-related (children cannot perform WRS). This is Missing Not At Random, meaning standard imputation methods are biased. Solution: request CAP/SIR for children.
2. **Sample size:** After SSD exclusion and requiring outcome, N=326 for adults. With 17 features, this is marginal (19 observations per feature).
3. **Imputation:** Currently using iterative imputation. Future work could explore multiple imputation (MICE) with proper uncertainty propagation.
4. **Complete cases:** Only 49.5% of adults have all Bertschinger features + outcome. Results may not generalize to patients with missing data.